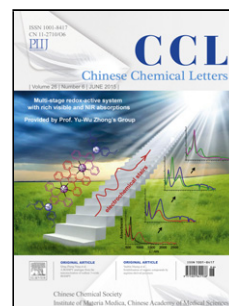


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Atomic Fe-N₄ sites on electrospun hierarchical porous carbon nanofibers as an efficient electrocatalyst for oxygen reduction reaction

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Communication

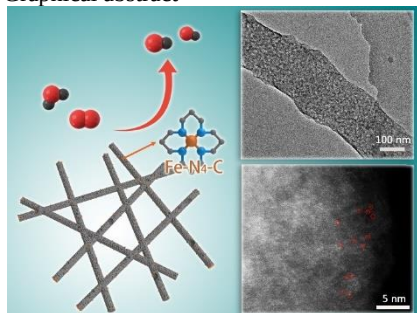
Atomic Fe-N₄ sites on electrospun hierarchical porous carbon nanofibers as an efficient electrocatalyst for oxygen reduction reactionRuiqin Zhong^{a,*,1}, Chenxu Zhi^{a,1}, Yingxiao Wu^{b,1}, Zibin Liang^b, Hassina Tabassum^b, Hao Zhang^b, Tianjie Qiu^b, Song Gao^b, Jinming Shi^b, Ruqiang Zou^{b,*}^a China Key Laboratory of Heavy Oil Processing, China University of Petroleum, Beijing 102249, China.^b Beijing Key Laboratory for Theory and Technology of Advanced Battery Materials, Department of Materials Science and Engineering, College of Engineering, Peking University, Beijing 100871, China.

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Graphical abstract



A novel hierarchical porous nitrogen-doped carbon nanofiber with atomically dispersed Fe-N₄ sites are prepared by pyrolyzed electrospun metal-organic frameworks /polyacrylonitrile nanofibers. One-dimension structure, abundant atomically dispersed metal sites and hierarchical pore structure of carbon nanofibers facilitates the electron transfer and enlarges the electrode-electrolyte interfaces. The carbon nanofibers exhibit superb electrocatalyst performance for oxygen reduction reaction, comparable to benchmark Pt/C catalysts in alkaline media.

ABSTRACT

Porous carbon materials doped with atomically dispersed metal sites (ADMSs) are promising electrocatalysts for oxygen reduction reaction (ORR) electrocatalysis. In this work, we fabricated hierarchical porous nitrogen-doped carbon nanofibers with atomically dispersed Fe-N₄ sites by carbonization of electrospinning iron-based metal organic frameworks (MOFs)/polyacrylonitrile nanofibers for ORR electrocatalysis. Remarkably, the resultant carbon nanofibers with atomically dispersed Fe-N₄ sites exhibit extraordinary electrochemical performance with an onset potential of 0.994 V and a halfwave potential of 0.876 V in alkaline electrolyte, comparable to the benchmark commercial Pt/C catalyst. The high catalytic performance is originated from the unique hierarchically porous 1D carbon structure and abundant highly active atomically dispersed Fe-N₄ sites.

Keywords: Atomically dispersed metal sites, Metal-organic frameworks, Electrospinning, Oxygen reduction reaction, Electrocatalysis

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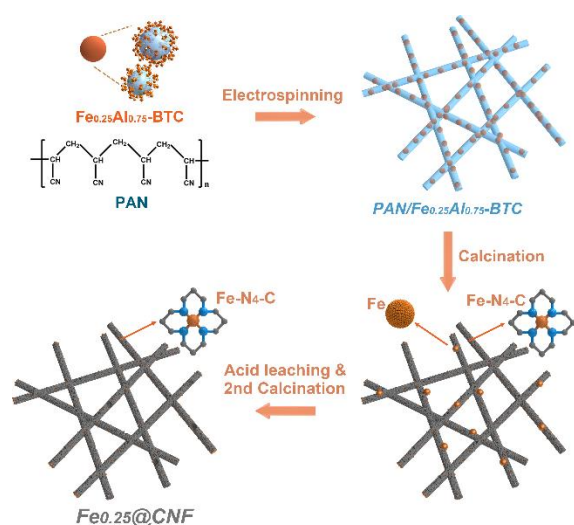
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Oxygen reduction reaction (ORR) is a key reaction for various energy conversion and storage systems, such as fuel cell and metal-air battery [1-3]. Their commercial application is limited by sluggish kinetics of ORR at the cathode, thus efficient catalysts are highly required. Up to now, the only commercialized Pt-based catalysts suffer from poor stability, crustal rarity, and prohibitive price. Hence, great efforts are devoted to developing non-precious metal catalysts to replace Pt-based catalysts. As one of the most promising candidates, transition metal (such as Fe and Co) atomically dispersed on nitrogen-doped carbon (M-N/C) catalysts exhibit superb ORR electrocatalytic activity and excellent stability [4-7]. Recently, single-atom catalysts are attracting increasing interests because of the presence of abundant highly active atomically dispersed metal sites (ADMS) [8-11]. Unique electronic structure and 100% atomic utilization of ADMSs lead to desirable electrochemical activity for ORR. However, the rational design of ideal ADMS-based carbon supports with expected structures and compositional features still remains a great challenge to date.

Electrospinning technique becomes a new research focus for energy conversion and storage field, which is a facile and scalable method to prepare carbon nanofibers [12-14]. The interpenetrating one-dimensional structure provides excellent passway for electron transportation and large surface for high expose of ADMSs. By structuring metal-organic frameworks (MOFs) into nanofibers, porous carbon nanofibers with well-dispersed ADMSs can be procured after carbonization [15, 16]. Fastened mass transport can be achieved by hierarchical pore structure bring by MOFs nanocrystals. Herein, we fabricated atomically Fe-N₄ sites dispersed hierarchical porous carbon nanofibers (Fe_{0.25}@CNF) carbonized from electrospun of polyacrylonitrile (PAN) nanofibers embedded with aerogel of Fe_{0.25}Al_{0.75}-BTC. The resulting carbon nanofibers own hierarchical pore structure, high N-doping level and well dispersed ADMSs active sites. The special characteristics make Fe_{0.25}@CNF exhibits high ORR activity with an onset potential of 0.994 V and a half-wave potential of 0.876 V, comparable to the benchmark commercial Pt/C catalyst.



Scheme 1 Schematic illustration of the synthetic process for Fe_{0.25}@CNF.

The synthesis procedure for Fe_x@CNF is schematically depicted in Scheme 1. First, Fe_xAl_{1-x}-BTC metal-organic aerogels (MOAs) were synthesized through the previously reported method [17, 18]. Due to poor crystallinity, the 3D microlites networks Fe_xAl_{1-x}-BTC-MOAs possess weak diffraction peaks in powder X-ray diffraction (PXRD) patterns (Figs. S1 and S2a in Supporting information). Then, Fe_xAl_{1-x}-BTC-MOAs were embedded into PAN nanofibers by a facile single-nozzle electrospinning method, which was support and carbon source to carbon nanofibers. For the purpose of conforming MOAs' dispersion, PXRD, scanning electron microscopic (SEM), and transmission electron microscopic (TEM) characterizations were performed. The diffraction peaks of Fe_xAl_{1-x}-BTC-MOAs could be identified in the PXRD patterns of Fe_xAl_{1-x}-BTC-PAN (Fig. S2b in Supporting information). Both SEM and TEM images illustrated the homogeneously loading of aerogels and the continuity and uniformity of MOA-PAN nanofibers precursor with average diameters of *ca.* 200 nm (Figs. S3 and S4 in Supporting information). During pyrolysis at 900 °C under an argon atmosphere, Fe³⁺ ions would be reduced by adjacent carbon species, and some of them were trapped by N and form Fe-N_x clusters through the Lewis acid-base interactions [13,19]. The MOA-PAN precursors transformed into N-doped carbon nanofibers with atomically dispersed Fe-N_x sites. Extra Fe and Al species were removed by acid leaching, forming abundant pore structures. The second calcination is necessary to improve the graphitization of carbon and remove unstable functional groups bring by acid etching. For comparison, Fe_xAl_{1-x}-BTC MOAs were treated by the same processes obtaining porous carbon materials named Fe_x/C.

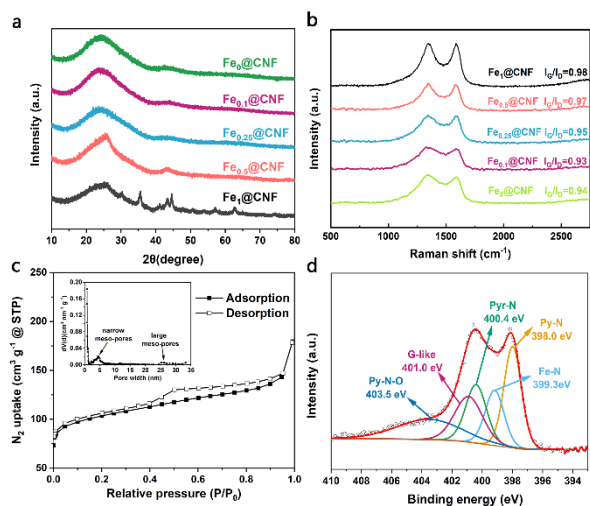


Fig. 1 (a) PXRD patterns of $\text{Fe}_x\text{@CNF}$ catalysts. (b) Raman spectra of $\text{Fe}_x\text{@CNF}$ catalysts. (c) N_2 sorption isotherms at 77 K for $\text{Fe}_{0.25}\text{@CNF}$, inset shows the corresponding pore size distributions. (d) High-resolution N 1s XPS spectrum of $\text{Fe}_{0.25}\text{@CNF}$.

The broad peaks at $\approx 26^\circ$ in PXRD patterns of $\text{Fe}_x\text{@CNF}$ corresponded to the (002) plane of graphitic carbon, and the diffraction peaks shifted to 26.5° , indicating more formation of graphitized carbon with increasing of iron content (Fig. 1a). Additionally, diffraction peaks of $\text{Fe}_1\text{@CNF}$ are indexed to Fe_3C (JCPDS No. 89-2867) phase. The graphitization degree information of $\text{Fe}_x\text{@CNF}$ could be further provided by Raman spectra (Fig. 1b) [20]. The intensity ratios of the tangential stretch G band ($\approx 1350\text{ cm}^{-1}$) to the disorder-induced D band ($\approx 1580\text{ cm}^{-1}$) for $\text{Fe}_x\text{@CNF}$ increase from 0.94 to 0.98 with the increase of Fe^{3+} molar ratio from 0 to 1. The surface areas and pore characteristics of all samples were characterized by 77 K isothermal N_2 sorption measurements (Fig. 1c, Figs. S5 and S6 in Supporting information). N_2 sorption curves of $\text{Fe}_{0.25}\text{@CNF}$ showed type-IV isotherms with obvious hysteresis loops indicating the existence of mesopores. The specific Brunauere Emmette Teller (BET) surface area and pore size distribution (PSD) of $\text{Fe}_x\text{@CNF}$ were summarized in Table S1 (Supporting information). The BET surface area of $\text{Fe}_{0.25}\text{@CNF}$ was calculated to be $386\text{ m}^2/\text{g}$. PSD revealed that micro- and mesopores constructed hierarchical structures in nanofibers (Fig. 1c insert). The full X-ray photoelectron spectroscopic (XPS) spectrum revealed surface nitrogen content in $\text{Fe}_{0.25}\text{@CNF}$ was 4.99 at% (Fig. S7 in Supporting information). The total nitrogen of $\text{Fe}_{0.25}\text{@CNF}$ was 8.51 wt% revealed by element analysis (EA) (Table S2 in Supporting information). Inductively coupled plasma-atomic emission spectroscopy (ICP-AES) analysis indicated that the Fe content of $\text{Fe}_{0.25}\text{@CNF}$ was 0.18 wt%, while no significant Fe 2p peak could be detected due to low content (Fig. S7 in Supporting information). As shown in Fig. 1d, high-resolution spectra of N 1s could be deconvoluted into five peaks with binding energies of 398.0, 399.3, 400.4, 401.0 and 403.5 eV corresponded to pyridinic N (Py-N), iron-coordinated (Fe-N), pyrrolic (Pyr-N), graphitic-like (G-like), and oxidized type (Py-N-O) nitrogen, respectively (Fig. 1d) [5].

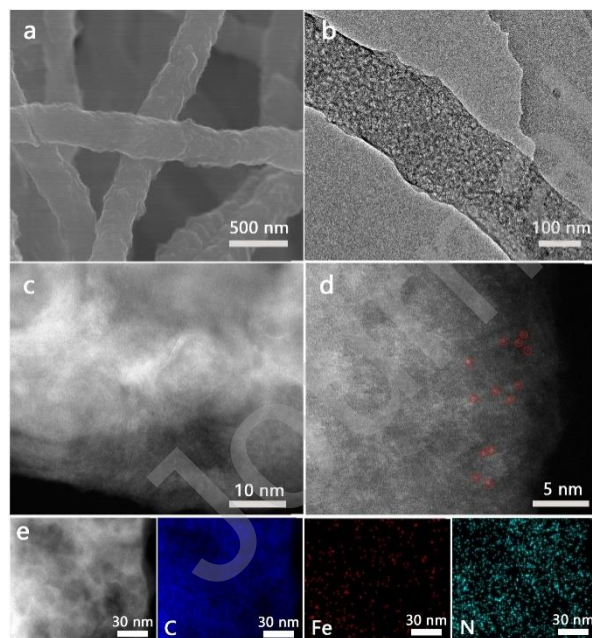


Fig. 2 (a) SEM, (b) TEM, (c, d) Aberration-corrected HAADF-STEM images and (e) corresponding EDS elemental mappings of $\text{Fe}_{0.25}\text{@CNF}$.

SEM and TEM images revealed that $\text{Fe}_{0.25}\text{@CNF}$ consists of uniform and continuous nanofibers with average diameters of *ca.* 200 nm (Figs. 2a and b). Lots of uniformly pore structures distributed on the surface and inside of the fibers, which attribute to iron particles removed by acid etching. Other four $\text{Fe}_x\text{/C}$ samples possessed metal bulks and nonporous carbon, which impeded the accessibility of active sites (Fig. S10 in Supporting information). As shown in TEM images of $\text{Fe}_{0.25}\text{@CNF}$, there are no crystalline iron nanoparticles or Fe clusters are observed, this provides the significant evidence that the iron has deposited in single-atom sites rather than aggregating into large assemblies. To confirm the existence of ADMs in $\text{Fe}_{0.25}\text{@CNF}$, atomic-resolution high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) was performed (Figs. 2c and d). Numerous isolated bright dots indicated the

existence of Fe-N₄ ADMs, and some of them marked with red cycles in Fig. 2d. Energy-dispersive X-ray spectroscopy (EDS) mapping analysis indicated the uniform elemental distribution of C, N, and Fe in Fe_{0.25}@CNF (Fig. 2e).

X-ray absorption near-edge structure (XANES) and extended X-ray absorption fine structure (EXAFS) analysis were performed to investigate the oxidation states and local coordination environment of Fe_{0.25}@CNF (Fig. 3). The Fe K-edge XANES showed that Fe oxidation state was close to Fe^{II} as in FePc. The magnitude of Fourier transformed k^2 -weighted Fe K-edge EXAFS spectrum exhibited the main peak at 1.44 Å. Using FePc as basic structure model, the fitting result proves that the main peak was assigned to Fe-N₁ coordination in the first shell. The second peak around 2.3 Å was assigned to Fe-C₁/C₂ coordination in the second shell. The absence of metallic Fe-Fe coordination certified that Fe sites were atomically dispersed in Fe_{0.25}@CNF.

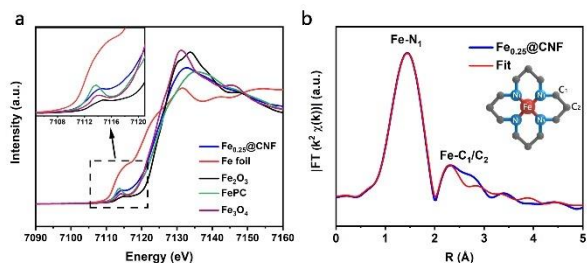


Fig. 3 (a) Fe K-edge XANES spectra of Fe_{0.25}@CNF, Fe foil, Fe₂O₃, FePc and Fe₃O₄. Inset shows the Fe K-edge pre-edge XANES. (b) Magnitude of Fourier transformed Fe K-edge EXAFS spectrum and the corresponding fitting curve of Fe_{0.25}@CNF.

In consideration of the advantages of single atomic Fe-N₄ active sites well-distributed in abundant pores, the Fe_{0.25}@CNF was considered to be an efficient catalyst for the oxygen reduction reaction with high stability. The performance of Fe_x@CNF was measured by cyclic voltammetry and linear sweep voltammetry in a 0.1 mol/L KOH solution using a three-electrode configuration. Fig. 4a reveals that a well-defined cathodic peak with a positive peak potential for Fe_{0.25}@CNF was measured in O₂-saturated electrolyte, which implied the electrocatalytic activity towards ORR. The ORR performance of Fe_x@CNF was further investigated by linear sweep voltammetry (LSV) using rotating-disk electrode (RDE). As shown in Fig. 4b and Table S3 (Supporting information), the best ORR catalytic performance was achieved by Fe_{0.25}@CNF with a positive onset potential of 0.994 V and half-wave potential of 0.876 V, which close to that of the commercial Pt/C catalyst.

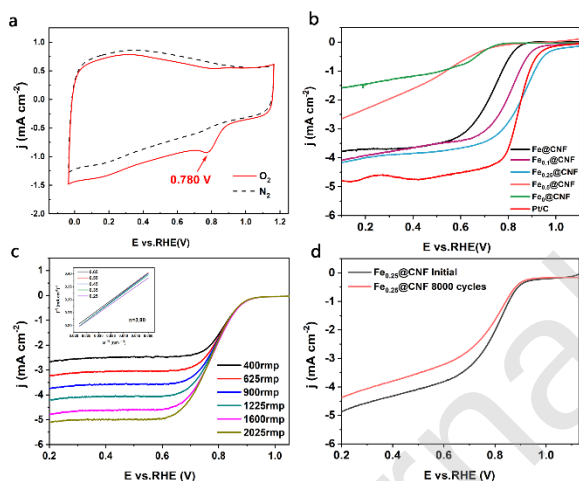


Fig. 4 (a) CVs of Fe_{0.25}@CNF in O₂-saturated 0.1 mol/L aqueous KOH (scan rate: 50 mV/s). (b) ORR polarization curves of Fe_x@CNF and Pt/C catalysts in O₂-saturated 0.1 mol/L KOH (scan rate: 10 mV/s, rotating speed: 1600 rpm). (c) ORR polarization curves of Fe_{0.25}@CNF at various rotating speeds in O₂-saturated 0.1 mol/L KOH (scan rate: 10 mV/s), inset shows corresponding Koutecky-Levich plots at various potentials. (d) ORR activity of Fe_{0.25}@CNF before and after durability tests.

For further insight into the reaction mechanism, the kinetic parameters including electron transfer number (n) and kinetic-limiting current density (j_k) were collected based on the polarization curves at various rotating speeds and corresponding Koutecky-Levich (K-L) plots (Fig. 4c). The electron transfer number for the Fe_{0.25}@CNF sample provided direct evidence of the catalyst performance. The K-L plots displayed excellent linearity and near coincidence, indicating similar electron transfer number of per oxygen molecule in ORR at different potentials. The electron transfer number was calculated to be 3.90 for Fe_{0.25}@CNF over the potential ranging from 0.25 V to 0.65 V, suggesting a preferred efficient four-electron pathway.

The long-term stability of Fe_{0.25}@CNF was tested by accelerated durability tests by cycling the electrode potential between 0.6 V and 1.0 V for 8000 cycles. Fe_{0.25}@CNF shown only a slight activity loss by ca. 20 mV negative shift in half-wave potential (Fig. 4d). Chronoamperometric measurements were employed to evaluate the long-term stability of Fe_{0.25}@CNF during ORR process, showing a low attenuation with high retention of 62% over 30000 s (Fig. S12 in Supporting information). The outstanding stability of Fe_{0.25}@CNF might be attributed to the strong interaction between Fe-N₄ ADMs and nitrogen-doped carbon substrates, which is consistent with reported works [21-23]. Furthermore, methanol crossover test showed that after addition of methanol into the electrolyte solution, no noticeable disturbance of the current density was observed for Fe_{0.25}@CNF, whereas the Pt/C catalyst exhibited a dramatic change in the current density due to the methanol oxidation (Fig. S13 in Supporting information). These results indicated that Fe_{0.25}@CNF possesses better electrocatalytic performance and durability than the commercial Pt/C catalyst for ORR in alkaline solutions.

The superb electrocatalytic performance of Fe_{0.25}@CNF could be attributed to the following key aspects: (1) The 1D structure of Fe_{0.25}@CNF provides a large exposed surface area and high electrical conductivity, which greatly enhances the electron-transfer

capability and promoting the electrocatalytic performance for ORR. (2) Hierarchical pore structure could enhance mass transportation, where macro- and mesopores work as a reservoir for an electrolyte, which shortens the diffusion distance. (3) The special structure of Fe_{0.25}@CNF creates extensive efficient Fe-N₄ sites that could actually participate in ORR reaction with ultrahigh current density and superb stability.

In summary, we prepared a hierarchical structure of Fe-N₄ sites well-dispersed porous carbon nanofibers via the thermal treatment of electrospun Fe_{0.25}Al_{0.75}-BTC@PAN composite precursor. The outstanding electrocatalytic performance of Fe_{0.25}@CNF for ORR could be attributed to the special structure of 1D carbon nanofibrous structure with high surface area and pore volume, the proper degree of the graphitization, and high content of the active species. The present results open up new avenues for developing novel open carbon nanostructures with controllable morphology and functionality for the next generation of nonprecious electrocatalysts.

Declaration of interests

The authors declare no conflict of interest.

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